

Student Name/ Number:



2006

YEAR 12

TRIAL HIGHERSCHOOL CERTIFICATE EXAMINATION

MATHEMATICS EXTENSION 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question

Total marks - 84

- Attempt Questions 1-7
- All questions are of equal value.

Total Marks – 84

Attempt Questions 1-7

All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your name and question number at the top of the page.

- | QUESTION 1. (12 MARKS) Begin a NEW sheet of writing paper. | Marks |
|--|-------|
| a) Solve $\frac{x}{x^2 - 4} < 0$ | 2 |
| b) For what values of x is $3^{2x} - (1 + \sqrt{3}) \times 3^x + \sqrt{3} = 0$? | 2 |
| c) Write the general solution to $\sqrt{3} \tan \theta - 1 = 0$ in exact radian form | 2 |
| d) Find $\int \sec^2 x \cdot \tan^2 x dx$ using the substitution $u = \tan x$ | 2 |
| e) Calculate the acute angle between the lines $2x - y - 1 = 0$
and $x - 2y + 1 = 0$ Give your answer to the nearest degree. | 2 |
| f) A parabola has the parametric equation $x = \sin \theta$, $y = \cos 2\theta$
What is the cartesian equation of this parabola? | 2 |

QUESTION 2 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

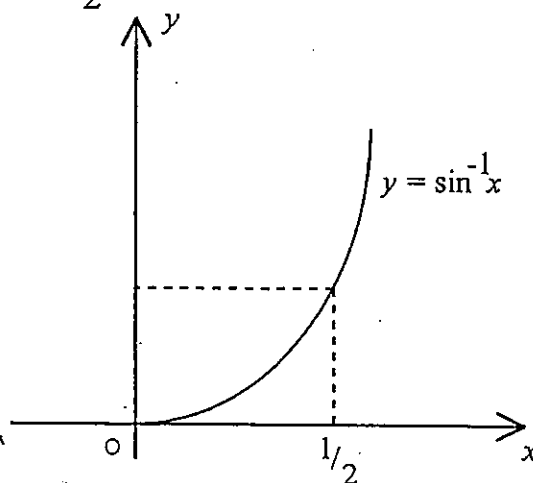
- a) With reference to the table of standard integrals find $\int \frac{\tan 2x}{\cos 2x} dx$ 2
- b) A particle is moving along the x axis. Its velocity V at position x is given by $V = \sqrt{8x - x^2}$. Find the acceleration when $x = 3$ 2
- c) On the same axes sketch the graphs of $y - 2x = 0$ and $y = -\cos x$ for $-\pi \leq x \leq \pi$. Use the graph to deduce the number of solutions to $2x + \cos x = 0$ 2
- d) Find the coordinates of the point which divides the interval joining $(3, -2)$ and $(-5, 4)$ *externally* in the ratio $5:2$ 2
- e) $(x - k)$ is a factor of $x^2 - 5x + (2k + 2)$. Find the value(s) of k . 2
- f) The equation $2x^3 + 12x^2 + 6x - 20 = 0$ has roots $\alpha - d, \alpha$ and $\alpha + d$. 1
- i) Find the value of α 1
- ii) Find a value of d . 1

QUESTION 3 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

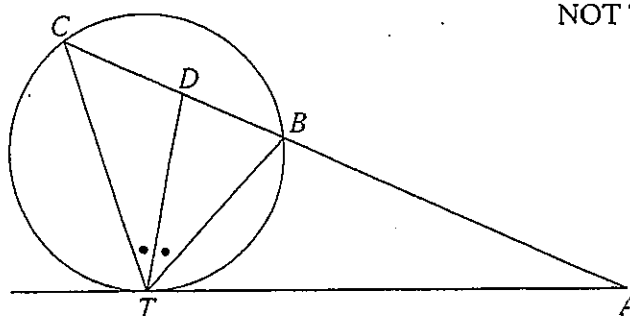
- a) i) Show that $x^3 + x^2 + x - 8 = 0$ has a root between $x = 1$ and $x = 2$. 1
- ii) Starting with $x = 2$ as the first approximation to the root of $x^3 + x^2 + x - 8 = 0$, use one application of Newton's method to find a better approximation to the root. 2

- b) Find the exact area bounded by the curve $y = \sin^{-1} x$, the x axis and the ordinate $x = \frac{1}{2}$, as shown in the diagram 3



- c) Differentiate $(x^2 + 2x + 2)e^{-x}$ and hence evaluate $\int_1^2 x^2 e^{-x} dx$ to 3 decimal places. 3

- d) TA is a tangent to a circle. Line ABCD intersects the circle at B and C. Line TD bisects $\angle BTC$. Prove $AT = AD$ 3

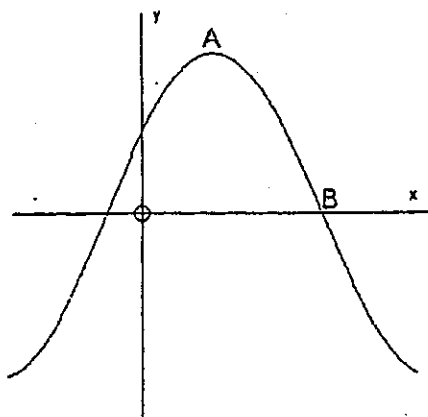


NOT TO SCALE

QUESTION 4 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

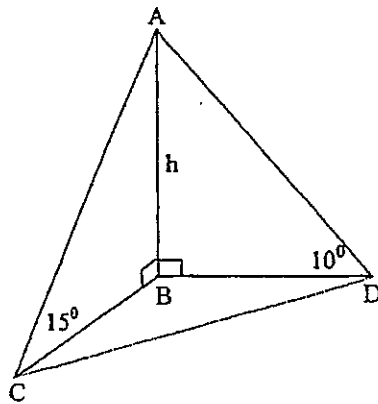
- a) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$
- Show the equation of the normal at P is $x + py = 2ap + ap^3$ 2
 - This normal cuts the y -axis at R . State the coordinates of R . 1
 - From P , a line PT is drawn perpendicular to the directrix, meeting it at T . State the coordinates of T . 1
 - If M is the midpoint of RT , find the coordinates of M . 1
 - Find the locus of M and show that it is a parabola with vertex at the focus of the original parabola. 2
- b)
- Express $\sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$; $R > 0$, $0 < \alpha < \frac{\pi}{2}$ 2
 - The graph of $y = \sqrt{3} \sin x + \cos x$ is shown here. Find the coordinates of A and B if A is a maximum turning point and B is where the curve cuts the x axis 3



QUESTION 5 (12 MARKS) Begin a NEW sheet of writing paper

- a) A B represents Kincumber mountain, height h metres. From points C and D in the same plane as the base of the mountain, the angles of elevation of the top of the mountain (A) are 15° and 10° respectively. From the base of the mountain, the bearings of the points C and D are 230° and 100° respectively.

- i) Find the size of angle CBD 1
- ii) Show $BD = h \cot 10^\circ$ 1
- iii) If CD is 450 metres find the height of the mountain 3



- b) i) Prove that $\frac{d}{dx} \left(\frac{x}{\sqrt{1-x^2}} \right) = \frac{1}{(1-x^2)^{\frac{3}{2}}}$ 2
- ii) Hence find the derivative of $\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right)$ 2
- iii) What restrictions are there on x 1
- iv) By considering a right angled triangle with a 1 unit hypotenuse show that $\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) = \sin^{-1} x$ for the domain $0 < x < 1$ 2

QUESTION 6 (12 MARKS) Begin a NEW sheet of writing paper.

- a) i) If $f(x) = 2 - \sqrt{x}$, $x \geq 0$ and $g(x) = (x - 2)^2$ for all x find the values of x for which $f[g(x)] = x = g[f(x)]$ 2
- ii) Find $f^{-1}(x)$ giving its domain. 1

- b) Use mathematical induction to prove that for all positive integers $n \geq 1$ 4

$$1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$$

- c) A particle is moving in a straight line with Simple Harmonic Motion. At time t seconds it has displacement x metres from a fixed point O, velocity v m/s and its acceleration a ms^{-2} is given by $a = -4x + 4$. Initially the particle is 2m to the right of O and moving away from O with speed $2\sqrt{3}ms^{-1}$

- i) Use integration to show $v^2 = -4x^2 + 8x + 12$ 2
- ii) Hence find the centre of the motion 1
- iii) If $x = 1 + 2\cos(2t + \alpha)$ for $0 < \alpha < 2\pi$ find the exact value of α . 2

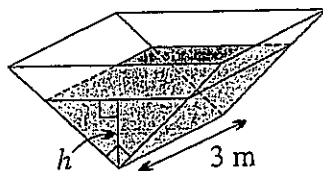
QUESTION 7 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

a) Whilst playing in the US Open Tennis Andre Agassi can serve a ball from the height of 1.8 metres. He hits the ball in a horizontal direction at a speed of 35m/s.

- i) Using $g = 10ms^{-2}$, derive expressions for the horizontal displacement x metres and the vertical displacement y metres, of the tennis ball after time t seconds of being hit. 2
- ii) Find how long before the ball hits the ground. 1
- iii) Find how far the ball will travel before bouncing. 1
- iv) By how much will the ball clear the net which is 0.95 metres high and 14 metres away from the service line. 2

- b) Due to council water restrictions a new tank in the shape of an isosceles triangular prism was installed. The tank was 3 metres long. A hose was used to fill the tank at a constant rate of 2 litres /second. The depth of water was h cm at time t seconds



- i) Find an expression for the volume of water in the tank (in cm^3 /second.) The depth of water is h cm 1
- ii) Find the rate at which the depth of the water is changing when $h = 20$ cm 2

QUESTION 7 CONTINUED

- c) A can of soft drink has an initial temperature of 18°C . To chill it Kim places it in her freezer that has a constant temperature of -19°C . The cooling rate of the soft drink is proportional to the difference between the temperature of the freezer and the temperature of the soft drink, T , that is $T = -19 + Ae^{-kt}$

- i) Find the value of A . **1**
- ii) After 5 minutes in the freezer the temperature of the drink is 3°C . Find the time it will take for the drink to reach a freezing temperature of 0° **2**

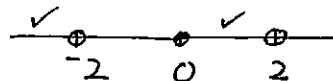
END OF EXAMINATION

Question 1.

EX 1 1 2006 TRIAL, GRS.

a) $\frac{x}{x^2-4} < 0$ c.p $x \neq \pm 2$

Solve $x = 0$



test $x = 1$

$\frac{1}{-3} \leq 0$ true

test $x = -3$

$\frac{-3}{5} \leq 0$ true

$x < -2$ $0 < x < 2$

b) let $m = 3^x$

but $3^x = m$

$m^2 - (1 + \sqrt{3})m + \sqrt{3} = 0$

$3^x = \sqrt{3}$ $3^x = 1$

$(m - \sqrt{3})(m - 1) = 0$

$x = \frac{1}{2}$ $x = 0$

$m = \sqrt{3}$ $m = 1$

c) $\sqrt{3} \tan \theta = 1$

$\tan \theta = \frac{1}{\sqrt{3}}$

$\theta = n\pi + \alpha$ in radians

$0 = n\pi + \frac{\pi}{6}$

d) $\int u^2 du$

$= \frac{u^3}{3} + c$

$= \frac{1}{3} \tan^3 x + c$

$u = \tan x$

$\frac{du}{dx} = \sec^2 x$

$du = \sec^2 x dx$

e) $2x - 1 = y$

$m_1 = 2$

$x + 1 = 2y$

$\frac{1}{2}x + \frac{1}{2} = y$

$m_2 = \frac{1}{2}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}} \right|$

$= \left| \frac{1\frac{1}{2}}{2} \right|$

$\theta = 37^\circ$

f) $x = \sin \theta$

$y = \cos 2\theta$

$= 1 - 2 \sin^2 \theta$

$= 1 - 2x^2$

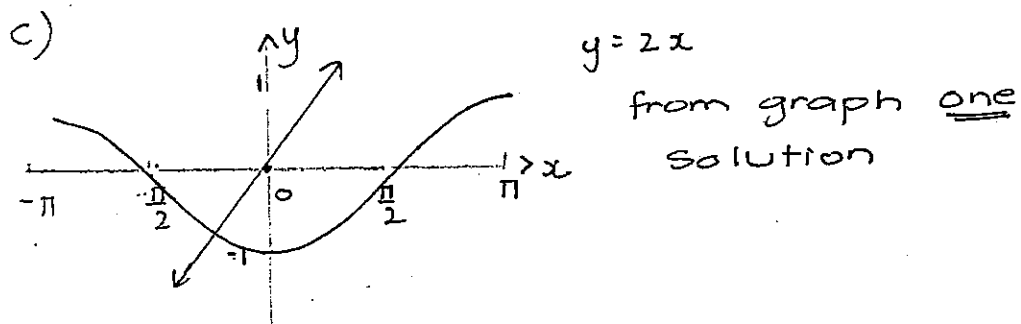
answer $y = 1 - 2x^2$

QUESTION 2.

a)
$$\int \frac{\tan 2x}{\cos 2x} dx = \int \sec 2x \tan 2x$$

$$= \frac{1}{2} \sec 2x + c$$

b) $v^2 = 8x - x^2$
 $\frac{1}{2} v^2 = 4x - \frac{x^2}{2}$
 $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4 - x$
 at $x = 3$ $\ddot{x} = 4 - 3$
 $= 1 \text{ m/s}^2$



d) $(3, -2)$ $(-5, 4)$
 $5 : 2$
 $x = \frac{6 - 25}{3}$ $y = \frac{4 - 20}{3} \Rightarrow x = \frac{-19}{3}$ $y = \frac{-16}{3}$

e) $f(x) = x^2 - 5x + (2k+2)$
 $f(k) = k^2 - 5k + 2k + 2 = 0$
 $k^2 - 3k + 2 = 0$
 $(k-2)(k-1) = 0$ $k=2$ or $k=1$.

f) $\alpha + \beta + \gamma = -\frac{b}{a}$ $\alpha \beta \gamma = -\frac{d}{a}$
 $a-d+a+a+d = -6$ $(a-d)(a)(a+d) = 10$
 $3a = -6$ $(-2-d) \times -2(-2+d) = 10$
 $a = -2$ $-8 + 2d^2 = 10$
 $2d^2 = 18$
 $d^2 = 9$
 $d = \pm 3$

QUESTION 3.

a) $f(x) = x^3 + x^2 + x - 8$

$$f(1) = -5$$

$$f(2) = 8 + 4 + 2 - 8 = 6$$

$f(1)$ and $f(2)$ have opposite signs and $f(x)$ is continuous

$\therefore$ a root exists between 1 & 2

ii) approx

$$= x_1 - \frac{f(x_1)}{f'(x_1)}$$
$$= 2 - \frac{6}{17}$$
$$= \underline{1.647}$$

$$f'(x) = 3x^2 + 2x + 1$$

$$f'(2) = 12 + 4 + 1 = 17$$

b) $y = \sin^{-1} x$ $x = \frac{1}{2}$ $x = 0$

$$y = \frac{\pi}{6} \quad y = 0$$

Area = rectangle - area to y axis

$$= \frac{1}{2} \times \frac{\pi}{6} - \int_0^{\frac{\pi}{6}} \sin y \, dy$$

$$= \frac{\pi}{12} - \left[-\cos y \right]_0^{\frac{\pi}{6}}$$

$$= \frac{\pi}{12} + \cos \frac{\pi}{6} - \cos 0$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$$

c) $\frac{d}{dx} \left[(x^2 + 2x + 2)e^{-x} \right]$

$$= (2x + 2) \cdot e^{-x} - e^{-x} (x^2 + 2x + 2)$$
$$= e^{-x} [2x + 2 - x^2 - 2x - 2]$$
$$= -x^2 e^{-x}$$

$$\int_1^2 x^2 e^{-x} \, dx = \left[-(x^2 + 2x + 2)e^{-x} \right]_1^2$$
$$= -e^{-2}(4 + 4 + 2) + e^{-1}(1 + 2 + 2)$$
$$= -10e^{-2} + 5e^{-1}$$
$$= 0.486$$

d) Let $\angle DTB = \beta$

$$\angle BTA = \angle TCB (\angle \text{in alternate segment}) = \alpha$$

$\angle TDA$ is exterior $\angle$ of $\triangle TCD$

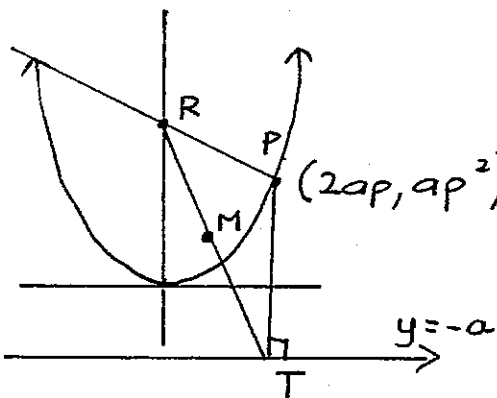
$$\therefore \angle TDA = \alpha + \beta$$

similarly $\angle DTA = \alpha + \beta$

$\therefore \triangle ADT$ is isosceles $\triangle$ (equal base $\angle$'s)

$$\therefore AT = AD$$

QUESTION . . 4



$$i) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} \text{ at } x = 2ap$$

$$m = p$$

$$\text{normal } m = -\frac{1}{p}$$

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

$$ii) y \text{ axis } x = 0 \quad py = 2ap + ap^3$$

$$(0, 2a + ap^2) = R.$$

$$iii) T \text{ directrix } y = -a$$

$$T(2ap, -a)$$

$$iv) R(0, 2a + ap^2) \quad T(2ap, -a)$$

$$M = (ap, \frac{a + ap^2}{2})$$

$$v) x = ap \quad y = \frac{a + ap^2}{2}$$

$$p = \frac{x}{a} \quad y = \frac{a}{2} + \left(\frac{x}{a}\right)^2 \cdot \frac{a}{2}$$

$$y = \frac{a^2 + x^2}{2a}$$

$$2ay - a^2 = x^2$$

$$2a(y - \frac{a}{2}) = x^2 \Rightarrow \text{vertex } (0, \frac{a}{2})$$

$$b) \sqrt{3} \sin x + \cos x = R \sin(x + \alpha)$$

$$R = 2$$

$$\frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x = \sin x \cos \alpha + \cos x \sin \alpha$$

$$\cos \alpha = \frac{\sqrt{3}}{2} \Rightarrow \alpha = \frac{\pi}{6} \quad (1 \text{ mark } \alpha)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin(x + \frac{\pi}{6})$$

$$ii) A = (\frac{\pi}{2} - \frac{\pi}{6}, 2)$$

$$= (\frac{\pi}{3}, 2)$$

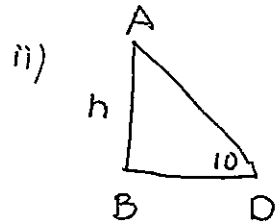
(1 mark each
for x and y)

$$B = \pi - \frac{\pi}{6}$$

$$= \frac{5\pi}{6}$$

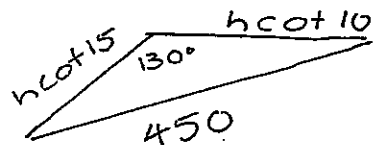
(1 mark for
 $B = \frac{5\pi}{6}$)

2) i) $\angle CBD = 130^\circ$ (1)



$$\begin{aligned}\tan 10^\circ &= \frac{h}{BD} \\ BD &= \frac{h}{\tan 10^\circ} \\ &= h \cdot \cot 10^\circ\end{aligned}$$

iii) $BC = h \cot 15^\circ$



$$450^2 = h^2 \cot^2 15 + h^2 \cot^2 10 - 2h^2 \cot 10 \cot 15 \cos 130^\circ$$

$$450^2 = h^2 [\cot^2 15 + \cot^2 10 - 2 \cot 10 \cot 15 \cos 130]$$

$$450^2 = h^2 \times 73.3 \dots \checkmark$$

$$h^2 = 2762.589$$

$$h = 52.56 \text{ m} \quad \checkmark$$

$$\begin{aligned} \text{b) } \frac{d}{dx} \left(\frac{x}{\sqrt{1-x^2}} \right) &= \frac{\sqrt{1-x^2} \cdot 1 - x \cdot \frac{1}{2} (1-x^2)^{-1/2} \cdot 2x}{(1-x^2)} \\ &= \frac{(1-x^2)^{1/2} + \frac{x^2}{(1-x^2)^{1/2}}}{(1-x^2)} \\ &= \frac{(1-x^2) + x^2}{(1-x^2)^{3/2}} \\ &= \frac{1}{(1-x^2)^{3/2}} \end{aligned}$$

ii) Note $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$

$$\frac{d}{dx} + \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$$

$$= \frac{1}{1 + \frac{x^2}{1-x^2}} \times \frac{1}{(1-x^2)^{3/2}}$$

$$= \frac{1-x^2}{1-x^2+x^2} \times \frac{1}{(1-x^2)^{3/2}}$$

$$= (1 - x^2) \times \frac{1}{(1 - x^2)^{3/2}}$$

$$= \frac{1}{\sqrt{1-x^2}}$$

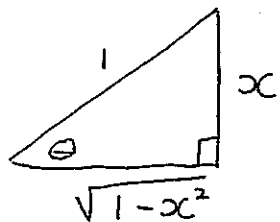
QUESTION 5 ctd.

b iii) $\tan^{-1} x$ exists for all real x

$\therefore \frac{x}{\sqrt{1-x^2}}$ must be real

$$\therefore -1 < x < 1$$

iv)



$$\sin \theta = \frac{x}{1}$$

$$\sin^{-1} x = \theta$$

$$\text{and } \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) = \theta$$

for $0 < x < 1$

$$\text{so } \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) = \theta = \sin^{-1} x$$

QUESTION 6

$$i). f(x) = 2 - \sqrt{x} \quad x \geq 0$$

$$g(x) = (x-2)^2$$

$$\begin{aligned} f[g(x)] &= 2 - \sqrt{(x-2)^2} \\ &= 2 - |x-2| \quad \text{for } x \leq 2 \\ &= 2 - -(x-2) \\ &= 2 + (x-2) \\ &= x \end{aligned}$$

$$\begin{aligned} g[f(x)] &= (2 - \sqrt{x} - 2)^2 \\ &= (-\sqrt{x})^2 \\ &= x \end{aligned}$$

$$\therefore f[g(x)] = g[f(x)] = x$$

for $x \geq 0$ and $x \leq 2$. or $0 \leq x \leq 2$

$$ii) y = 2 - \sqrt{x}$$

$$x = 2 - \sqrt{y}$$

$$\sqrt{y} = (2-x)$$

$$y = (2-x)^2$$

$$f^{-1}(x) = (2-x)^2$$

Domain $x \leq 2$ $y \geq 0$

$$b) 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$$

• prove true for $n=1$

$$\begin{aligned} \text{LHS} &= 1^2 \\ &= 1 \end{aligned} \quad \begin{aligned} \text{RHS} &= \frac{1}{3} \cdot 1 \cdot (2-1)(2+1) \\ &= 1 \end{aligned}$$

hence true for $n=1$

• assume true for $n=k$

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$$

• prove true for $n=k+1$ if true for $n=k$

$$\begin{aligned} &1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 \\ &\quad \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2 \\ &= \frac{1}{3}k(2k-1)(2k+1) + \frac{3}{3}(2k+1)^2 \\ &= \frac{1}{3}(2k+1) \{ k(2k-1) + 3(2k+1) \} \\ &= \frac{1}{3}(2k+1) \{ 2k^2 + 5k + 3 \} \\ &= \frac{1}{3}(2k+1) \{ k+1 \} \{ 2k+3 \} \\ &= \frac{1}{3}(k+1)(2(k+1)-1)(2(k+1)+1) \\ &= \text{RHS.} \end{aligned}$$

since it is true for $n=1$, true for $n=1+1$
i.e $n=2$, if it is true for $n=k$, then true
for, $n=k+1$ and so on for all positive
integers $n \geq 1$

NOTE

$$\sqrt{(x-2)^2} = \pm(x-2)$$

If $x > 2$

$$2 - (x-2) = 4-x$$

This is not needed
as told $f[g(x)] = x$

If $x \leq 2$

$$2 - -(x-2)$$

which is required.

QUESTION 6 (c)

$$\ddot{x} = -4x + 4$$

$$\frac{1}{2}v^2 = \int -4x + 4 dx$$

$$\frac{1}{2}v^2 = -2x^2 + 4x + C$$

$$\text{at } x = 2 \quad v = 2\sqrt{3}$$

$$6 = -8 + 8 + C$$

$$C = 6$$

$$\frac{1}{2}v^2 = -2x^2 + 4x + 6$$

$$v^2 = -4x^2 + 8x + 12$$

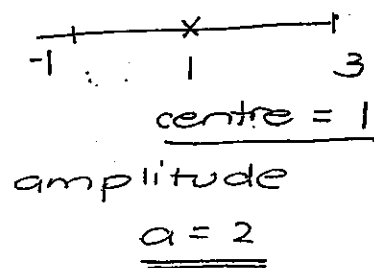
$$\text{ii) at } v = 0$$

$$4x^2 - 8x - 12 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = 3 \quad x = -1$$



$$\text{iii) when } t = 0 \quad x = 2$$

$$x = 1 + 2\cos(\alpha)$$

$$\therefore 2 = 1 + 2\cos(\alpha)$$

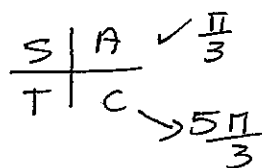
$$1 = 2\cos(\alpha)$$

$$\frac{1}{2} = \cos \alpha$$

$$\alpha = \frac{\pi}{3} \text{ or } \alpha = \frac{5\pi}{3}$$

check which one?

$$v = -4\sin(2t + \alpha)$$



$$\text{when } t = 0 \quad v = 2\sqrt{3}$$

$$2\sqrt{3} = -4\sin \alpha$$

$$\therefore \sin \alpha = -\frac{\sqrt{3}}{2}$$

$$\alpha = \frac{5\pi}{3} \text{ to satisfy both } \alpha \text{ and } v$$

S	A			
T	C			
✓	✓			

QUEST 10 N 7

$$\text{a) } \ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$\text{when } t = 0 \quad \dot{y} = v \sin \theta$$

$$\therefore c = v \sin \theta$$

$$c = 35 \sin \theta$$

$$= 0$$

$$\dot{y} = -10t$$

$$y = -5t^2 + c_1$$

$$\text{at } t = 0 \quad y = 1.8 \therefore c_1 = 1.8$$

$$y = -5t^2 + 1.8$$

Horizontal

$$\dot{x} = v \cos \theta$$

$$= 35 \cos \theta$$

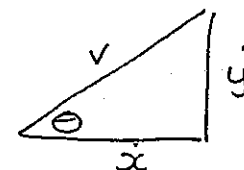
$$= 35$$

$$x = 35t$$

$$\text{ii) when } y = 0$$

$$0 = -5t^2 + 1.8$$

$$t = 0.6 \text{ sec.}$$



$$v \sin \theta = y$$

$$\cos \theta = \frac{x}{v}$$

$$v \cos \theta = \dot{x}$$

$$\text{iii) find } x \text{ at } t = 0.6$$

$$x = 35 \times 0.6$$

$$= 21 \text{ metres.}$$

$$\text{iv) Find } y \text{ when}$$

$$x = 14 \text{ find } t \text{ for}$$

$$14 = 35 \times t$$

$$t = 0.4 \text{ sec.}$$

$$\text{at } t=0.4$$

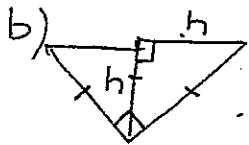
$$y = -5(0.4)^2 + 1.8$$

$$= 1 \text{ metre.}$$

∴ Ball clears net by

$$1 - 0.95 = .05 \text{ metre}$$

$$\text{or } 5 \text{ cm.}$$



$$\tan 45 = 1$$

∴ base is also h.

$$\text{Area} = \frac{1}{2} \times 2h \times h$$

$$= h^2$$

$$\text{Volume} = h^2 \times 300$$

$$= 300h^2 \text{ cm/sec}^3$$

$$\text{ii) } \frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$2000 = 600h \times \frac{dh}{dt}$$

$$\frac{2000}{600h} = \frac{dh}{dt}$$

$$\frac{20}{6 \times 20} = \frac{dh}{dt} \Rightarrow \frac{1}{6} \text{ cm/sec.}$$

NOTE
UNITS
h is cm
need 3m
as 300cm

$$\text{c) } T = -19 + Ae^{-kt}$$

$$\text{i) at } t=0 \quad T=18$$

$$18 = -19 + Ae^0$$

①

$$37 = A$$

$$\text{ii) } T = -19 + 37e^{-kt}$$

$$t=5 \quad T=3$$

$$3 = -19 + 37e^{-5k}$$

$$e^{-5k} = \frac{22}{37}$$

$$k = -\frac{1}{5} \log_e \frac{22}{37}$$

$$\approx 0.103975091$$

$$0 = -19 + 37e^{-kt}$$

$$\frac{19}{37} = e^{-kt}$$

$$\log_e \left(\frac{19}{37} \right) = -0.103975091t$$

$$t \approx 6.4 \text{ mins or } 6 \text{ min } 25 \text{ sec}$$